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An Introduction to Reliability Theory

4.1 Introduction

The study of reliability specification and performance requires a thorough understanding of many different concepts from reliability theory. Reliability theory deals with the interdisciplinary use of probability, statistics, and stochastic modelling, combined with engineering insights into the design and the scientific understanding of the failure mechanisms, to study the various aspects of reliability. It encompasses issues such as:

- Reliability modelling
- Reliability analysis and optimization
- Reliability engineering
- Reliability science
- Reliability technology
- Reliability management

In this chapter we briefly discuss these concepts that will be used in later chapters of the book. For readers who are familiar with reliability theory this chapter serves as a review chapter. For readers who are not familiar with reliability theory, we indicate references where they can get more details of the topic under consideration. The two references that are cited often are Blischke and Murthy (2000) and Rausand and Høyland (2004). Other references are indicated as and when appropriate.

The outline of the chapter is as follows: Section 4.1 defines basic concepts of reliability, like functions, failures, and failure modes and effects. Section 4.2 introduces reliability measures and lifetime models with focus on the exponential and Weibull models. System modelling by means of reliability block diagrams and fault tree analysis is outlined. How to incorporate environmental effects into life models, for example, by using proportional hazards models is also briefly discussed. Section 4.5 deals with modelling of repairable systems with both corrective and preventive maintenance strategies. Qualitative and quantitative reliability analyses are presented in Section 4.6 and reliability engineering issues are discussed in Section 4.7 with a special focus on reliability allocation and reliability growth. Sections 4.8 and 4.9

present a brief introduction to reliability prediction and to reliability management issues. The chapter concludes in Section 4.10 with a case study on cellular phones.

In the rest of this chapter we will use the term *item* to denote any physical entity, be it a large system or a small component.

4.2 Basic Concepts

In Section 1.3.1 we presented the definition of reliability from IEC 60050-191. The concept of reliability is related to one or more product functions that are required or wanted. Some functions are very important, while others may be of the category "nice to have". When we use the term reliability, we should always specify the required functions. The reliability of a product is dependent on the environmental and operational conditions during the product's post-production phase. These conditions have to be properly understood and assessed in order to develop a reliable product.

4.2.1 Product Functions

The key term in the definition of reliability is the ability of the item to perform a required *function*. The different functions of a complex item may be classified as follows (Rausand and Høyland, 2004).

Essential functions: These functions are the intended or primary functions, and may be considered as the reason why the item has been developed. The essential function of a pump is, for example, to pump fluid.

Auxiliary functions: These functions are required to support the essential function. An auxiliary function of a pump is, for example, to contain the fluid and prevent leakage to the environment.

Protective functions: These functions are intended to protect people, material assets, and the environment from damage, negative health effects, and injury.

Information functions: These functions give information from condition monitoring gauges, alarms, and so on.

Interface functions: These functions are related to the interfaces between the item considered and other items.

Superfluous functions: In some cases an item may have functions that are never used. This is sometimes the case with electronic equipment that has a wide range of "nice-to-have" functions that are often not necessary. In some cases, failure of a superfluous function may cause failure of a required function.

4.2.2 Failure and Related Concepts

Failure

A failure occurs when the item is not able to perform one or more of its required functions. Two definitions of failure are: